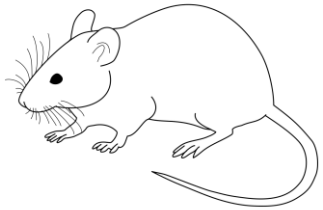
Estimation of Hemodynamic Response Function using Multimodal Recording – Linking EEG with HRF

Chi-Yuan Chang
Emily Stephen
David Boas
Michael Esterman

09/28/2026

Background





Neural Activity

Ca^{2+} Activity

EEG

HRF

Neuromodulation

Norepinephrine (NE)

Pupil size

HRF

Oxyhemoglobin (HbO)

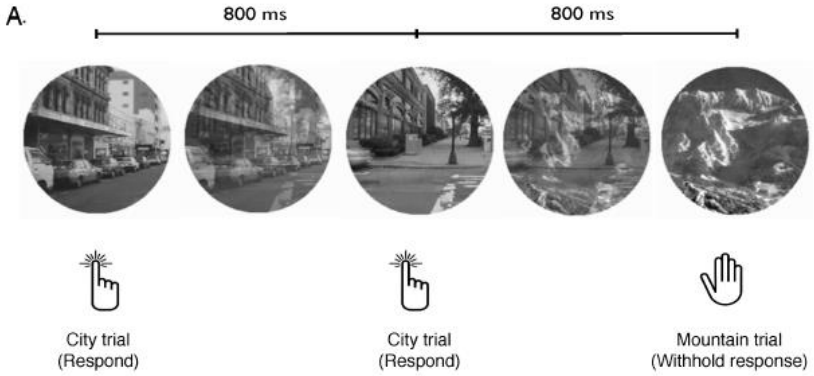


Optical imaging

fNIRS

Rauscher et al. 2025

GradCPT Experiment



6 mins Resting	6 mins Run 1	6 mins Run 2	6 mins Run 3

- Oddball: Mountain trial (~10%)
- Distractor: City trial (~90%)
- 450 trials per Run

Esterman et al. 2012

Modalities used:

- High-dense fNIRS
- 4 midline-channels EEG
- Eye tracker

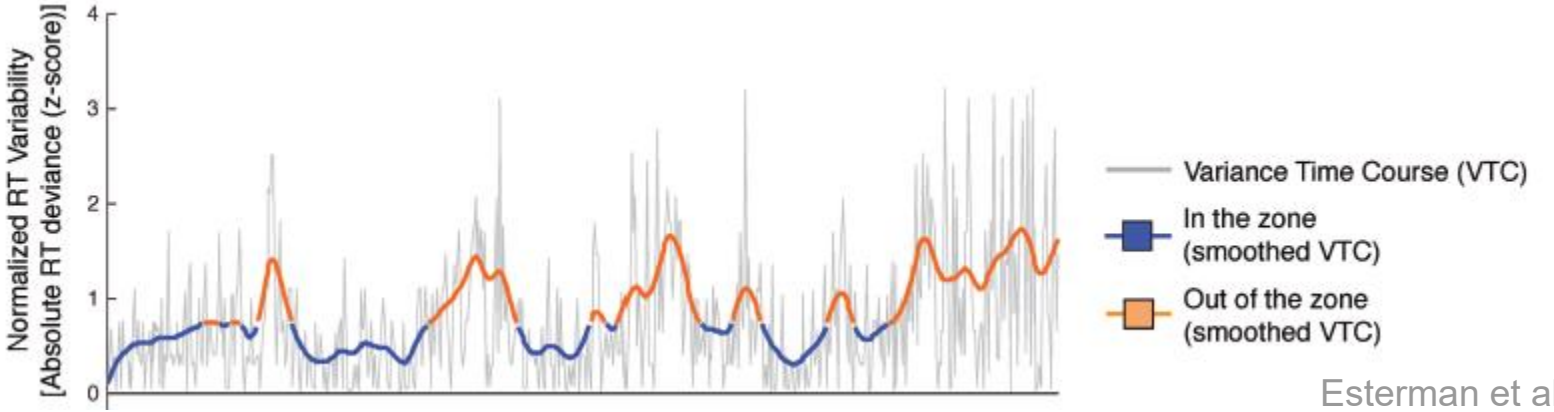


Variance Time Course (VTC)

- Quantify the inter-trial difference of Reaction Time.
- The more focus the subject is, the smaller the VTC is.

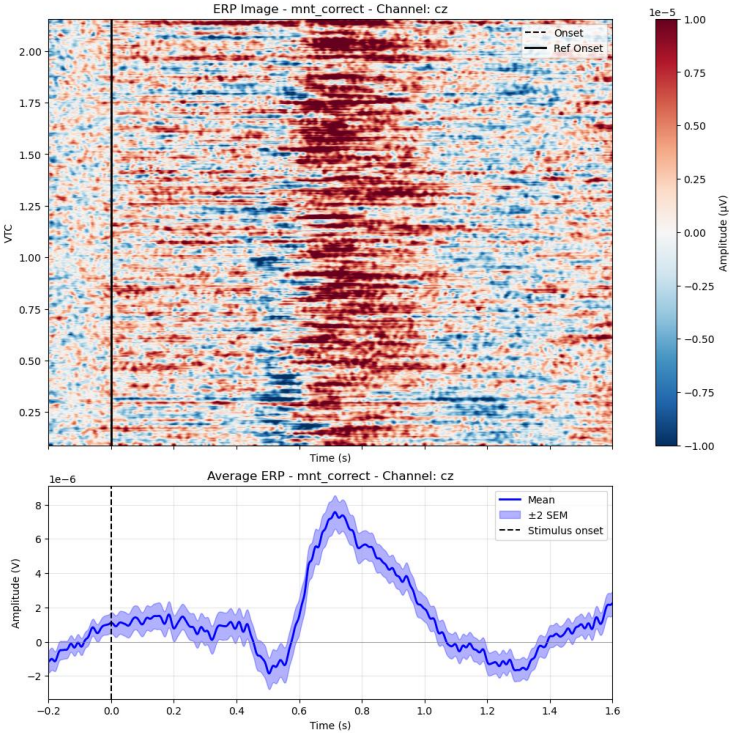
In the zone / Out of the zone

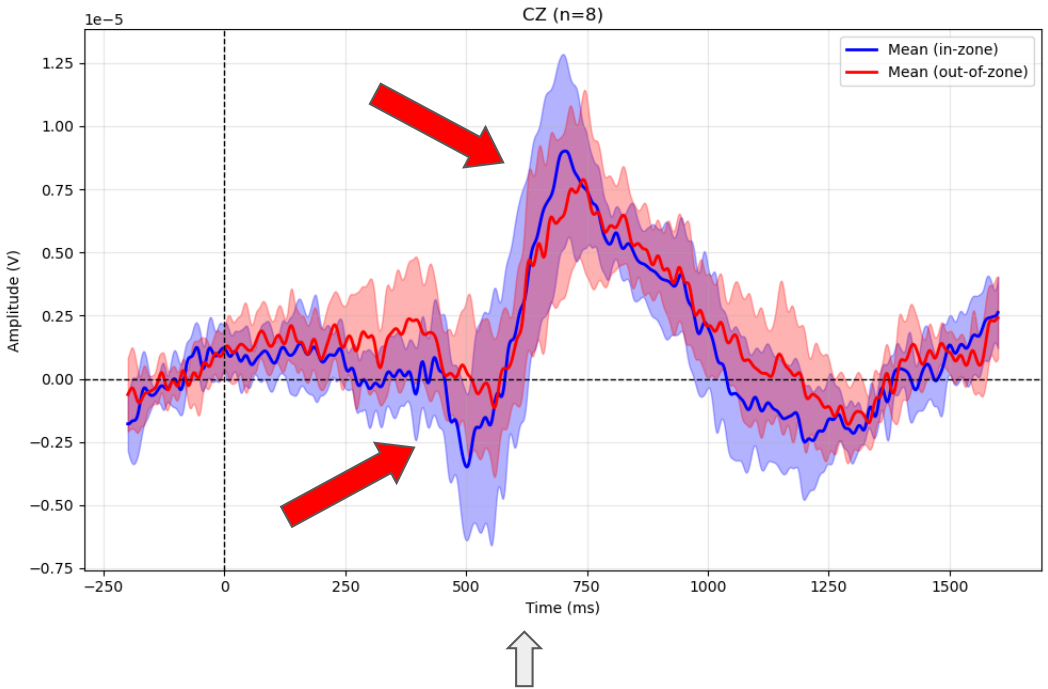
- In the zone: Duration when VTC is **lower** than the medium of VTC
- Out of the zone: Duration when VTC is **higher** than the medium of VTC



Esterman et al. 2012

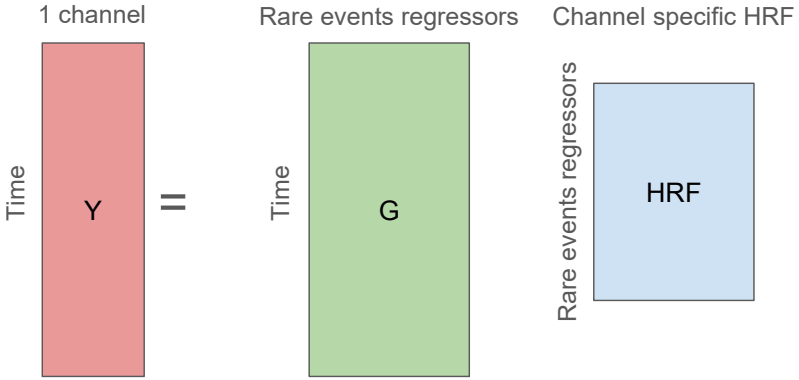
EEG analysis (ERP)



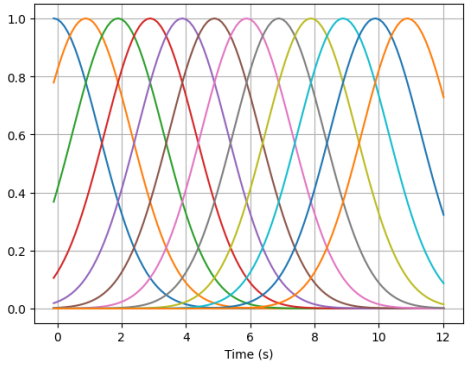


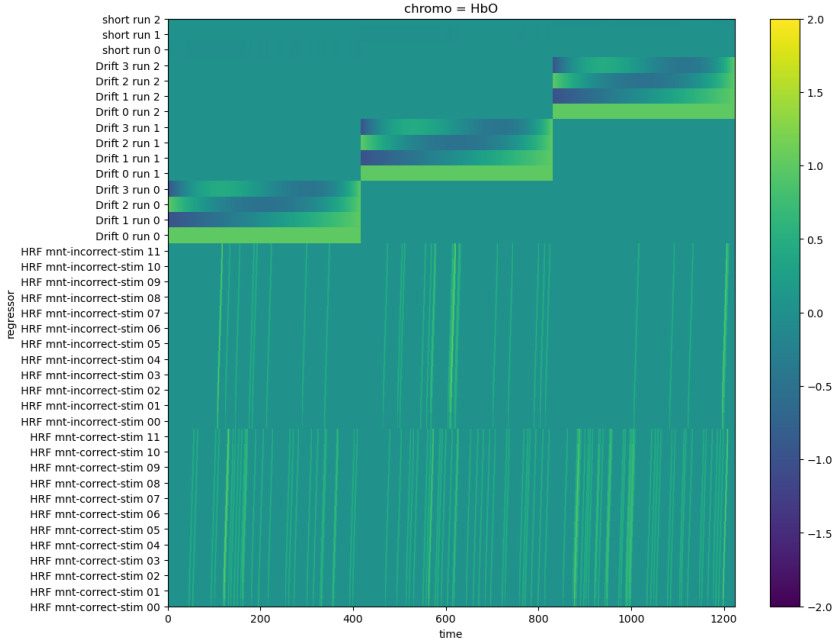
Clear N2 and P3 with a delay.

Previous GLM model

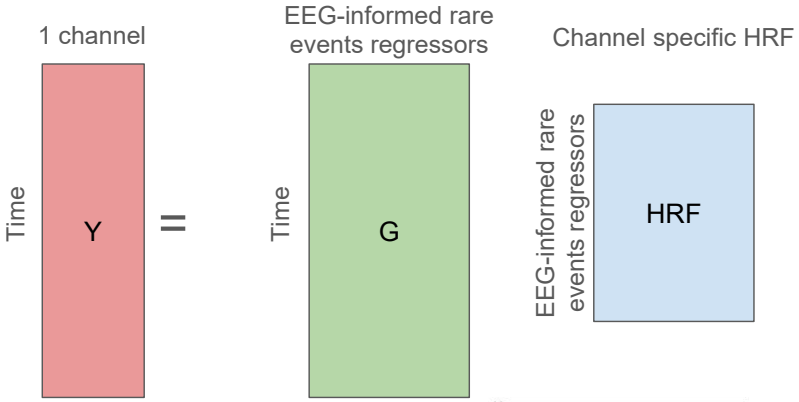


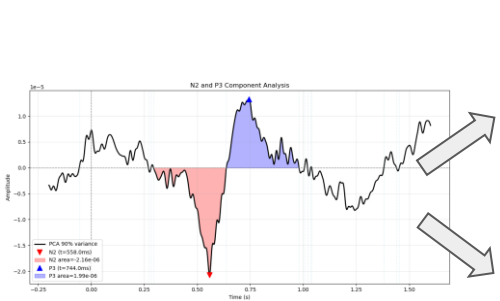
Consecutive gaussians

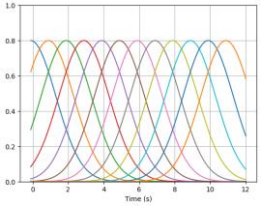


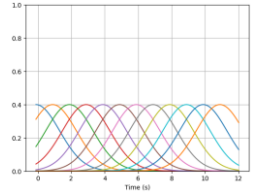


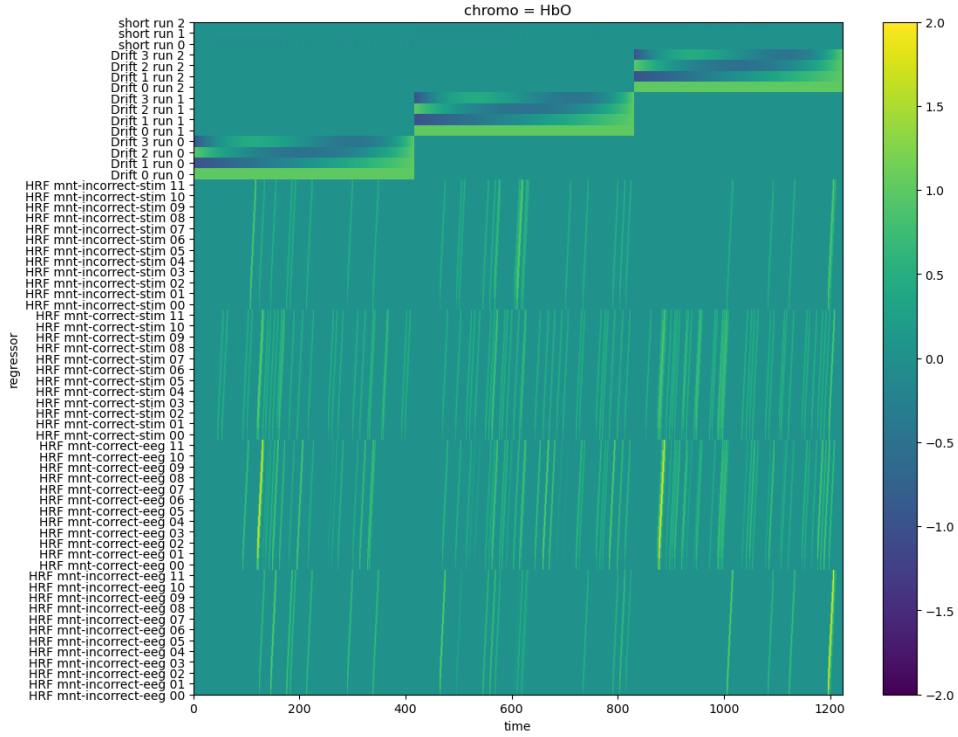
EEG-informed GLM model





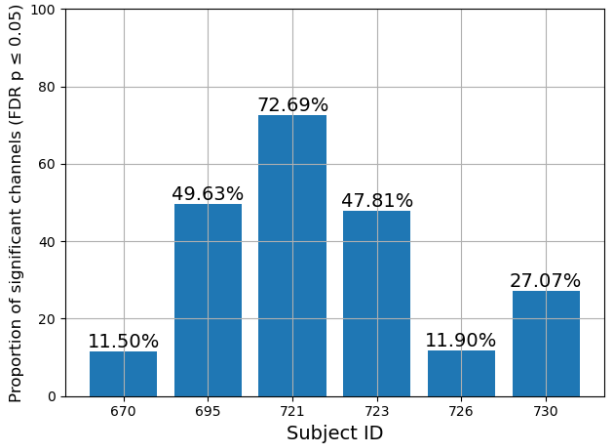




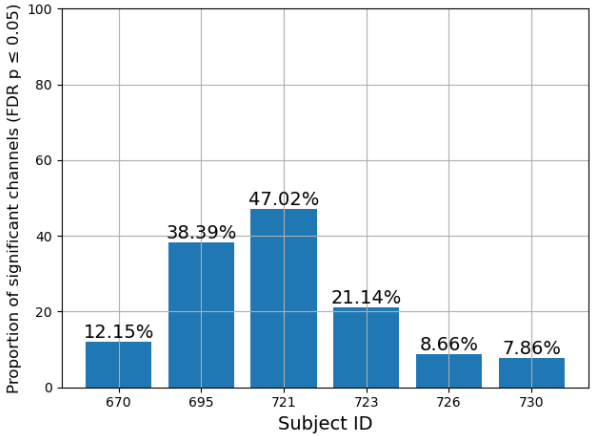


Proportion of Significant Channels (FDR corrected)

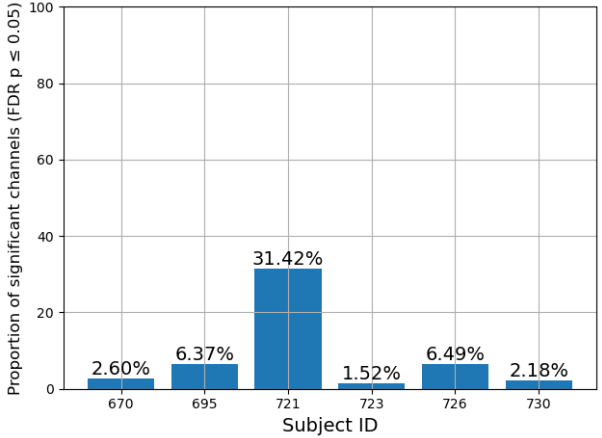
Full vs Basis



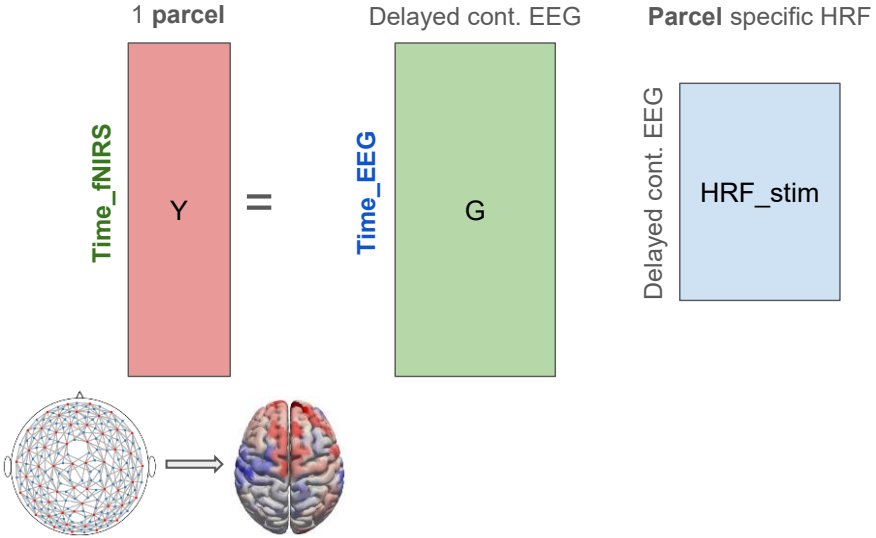
Reduced vs Basis



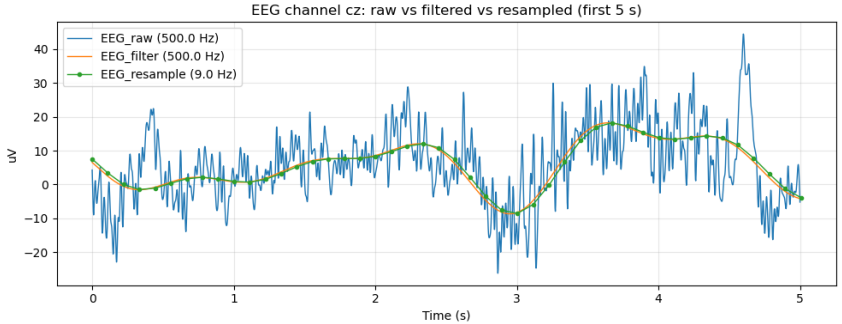
Full vs Reduced



Modeling HRF using continuous EEG



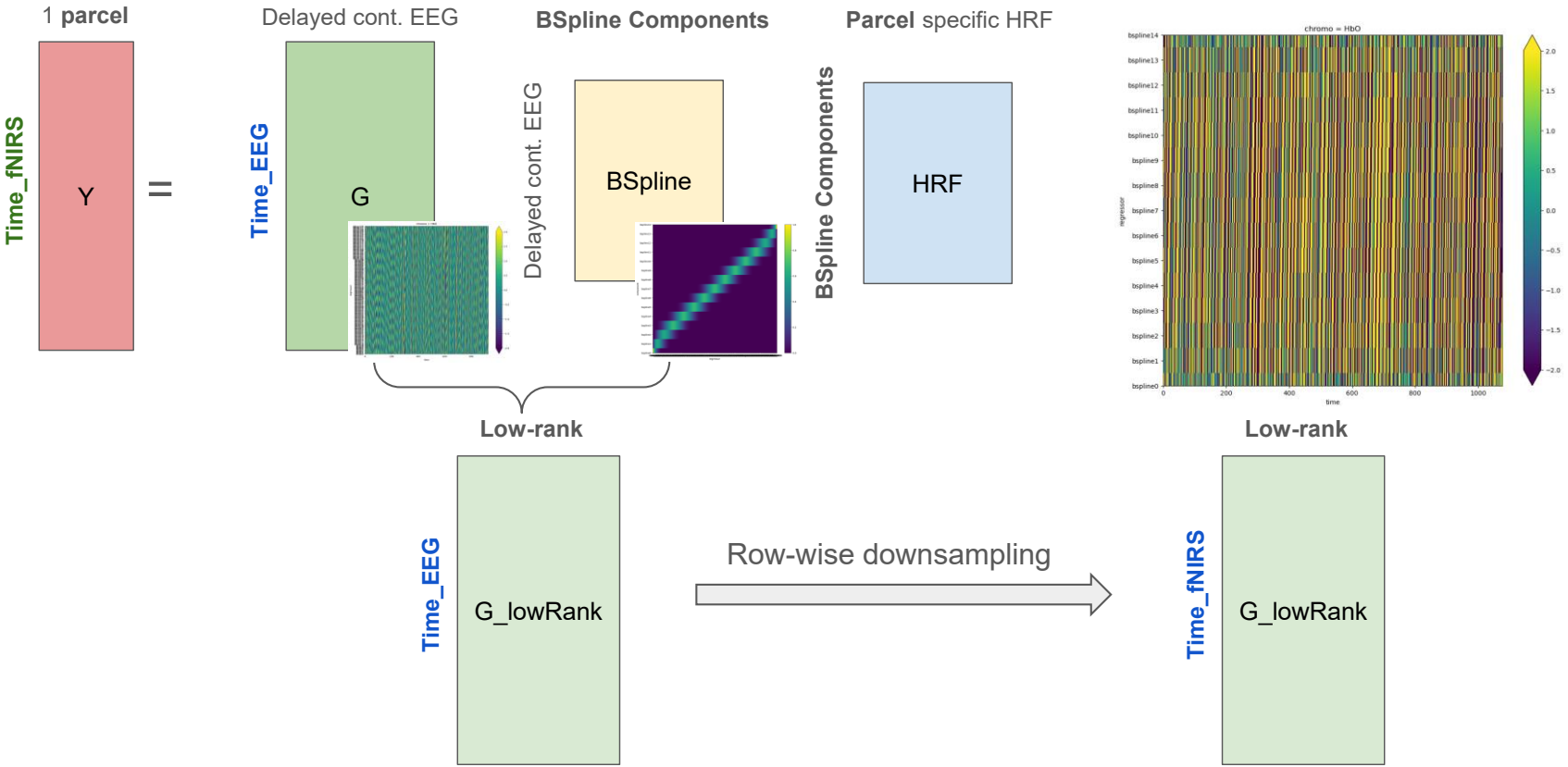
- fNIRS sfreq = 8.9Hz
- EEG sfreq = 500Hz
- 15 seconds delay results in 7500 EEG regressors



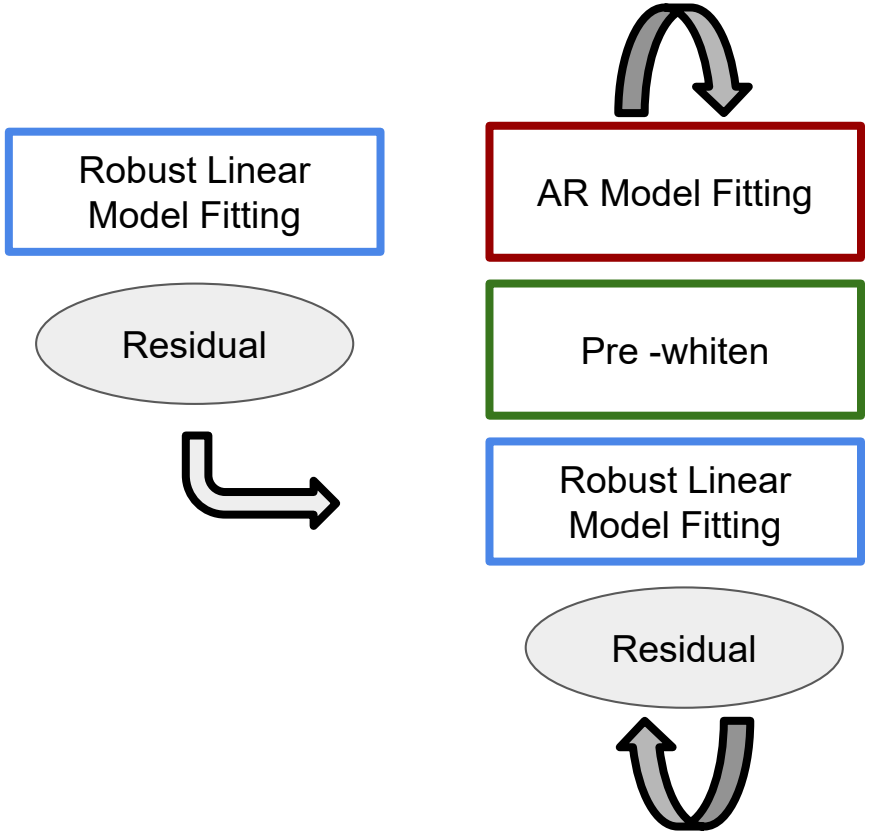
Two major questions:

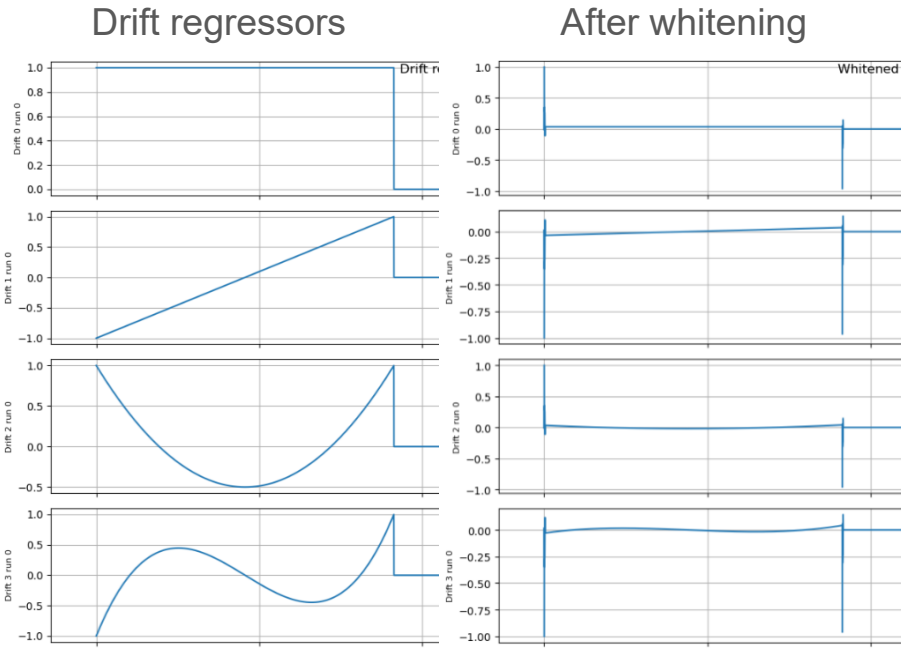
1. **Time_fNIRS** and **Time_EEG** have different length due to different sampling rate.
2. # of regressors (Delayed cont.EEG) are huge due to EEG's high sampling rate.

Low-rank representation of Delay using BSpline



Autoregressive Iterative Robust Least Square (AR-IRLS)



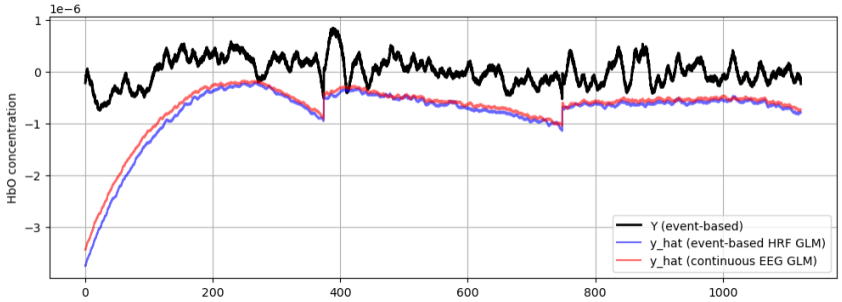


Barker et al. 2013

2-stage Model with high pass filter

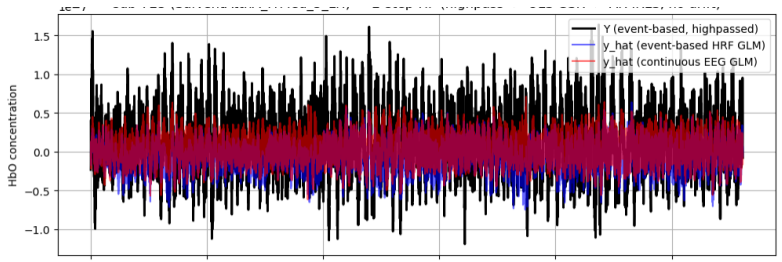
1-stage

Y = Drift GSR Other → AR-IRLS

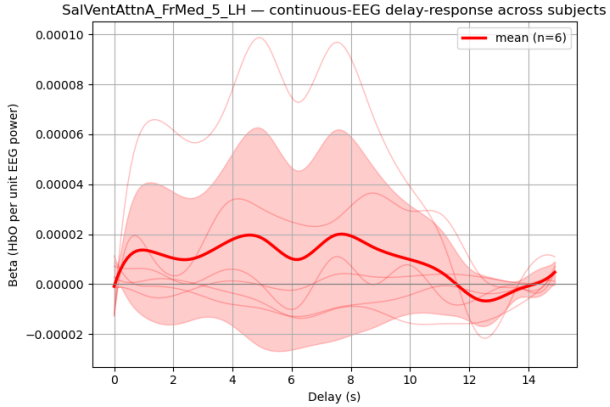


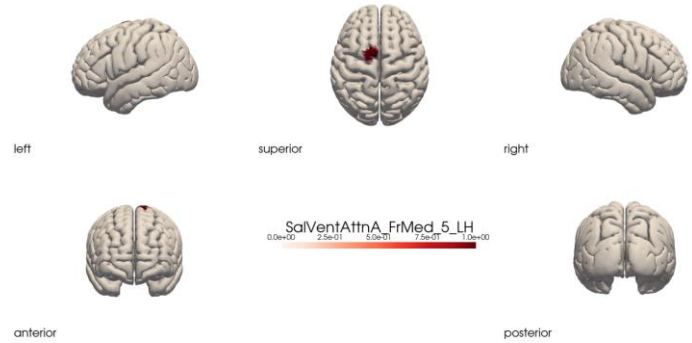
2-stage

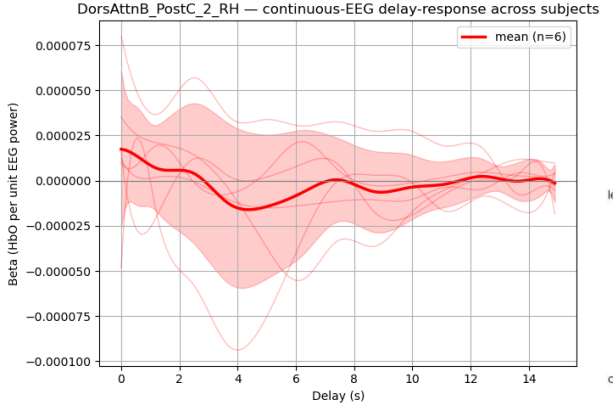
Y → HP → Y_hp = GSR Other (OLS) → AR-IRLS

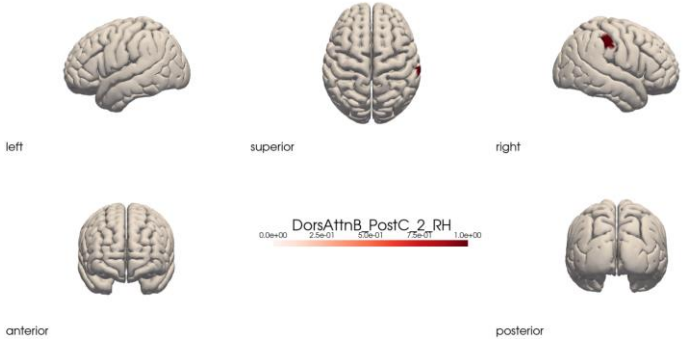


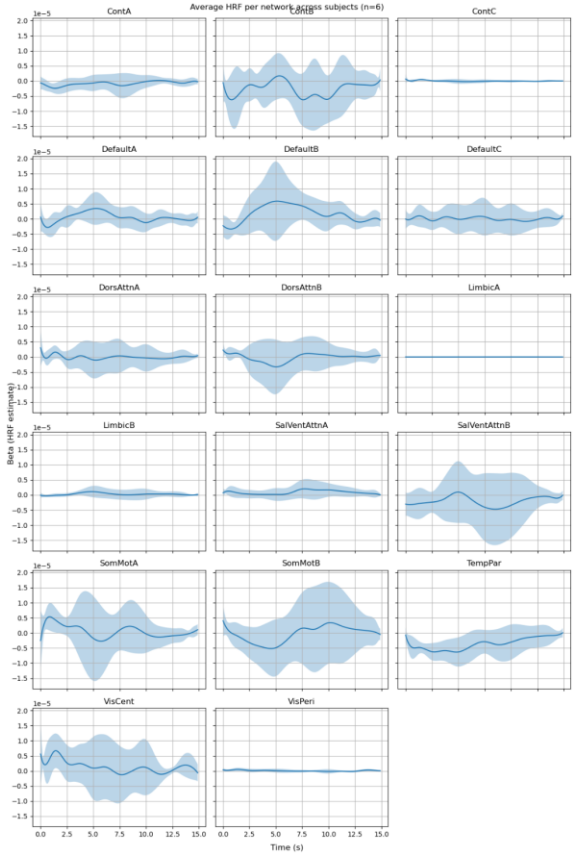
Example parcel HRF across subjects



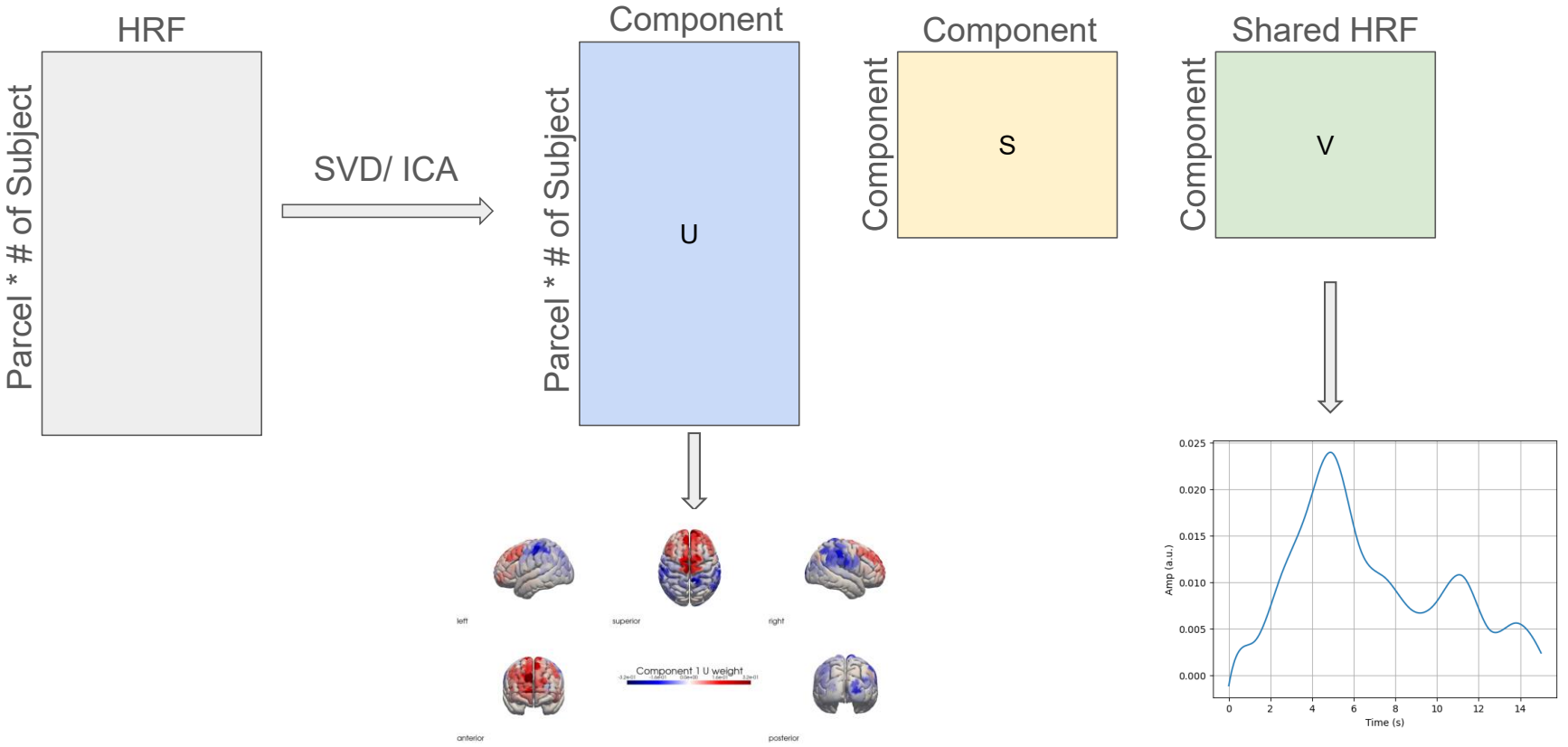




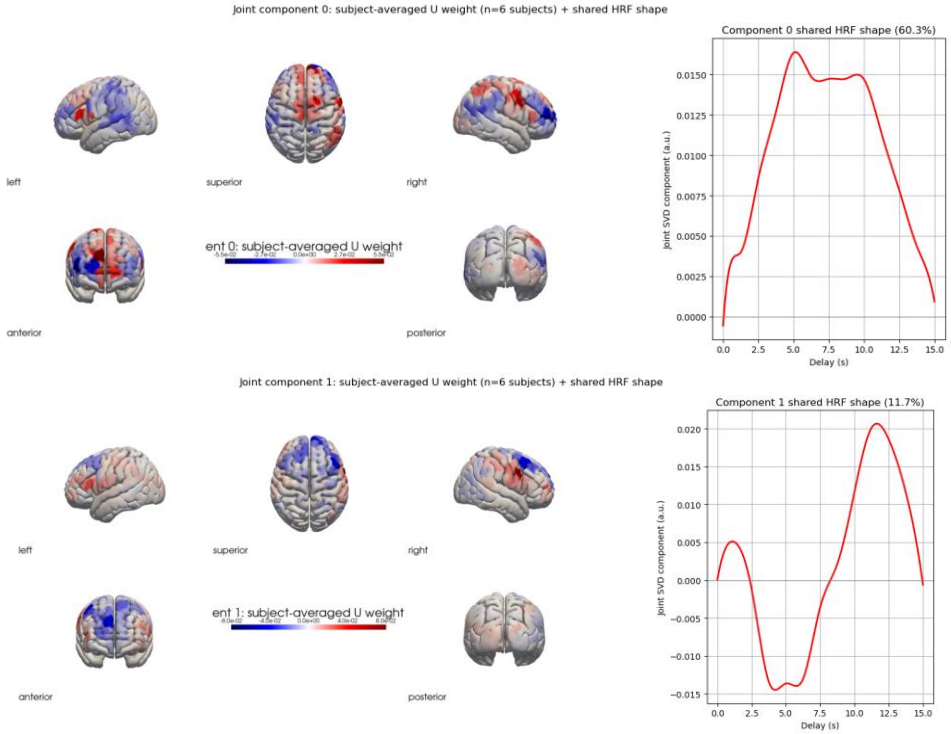




xSubject Parcel-wise shared HRF

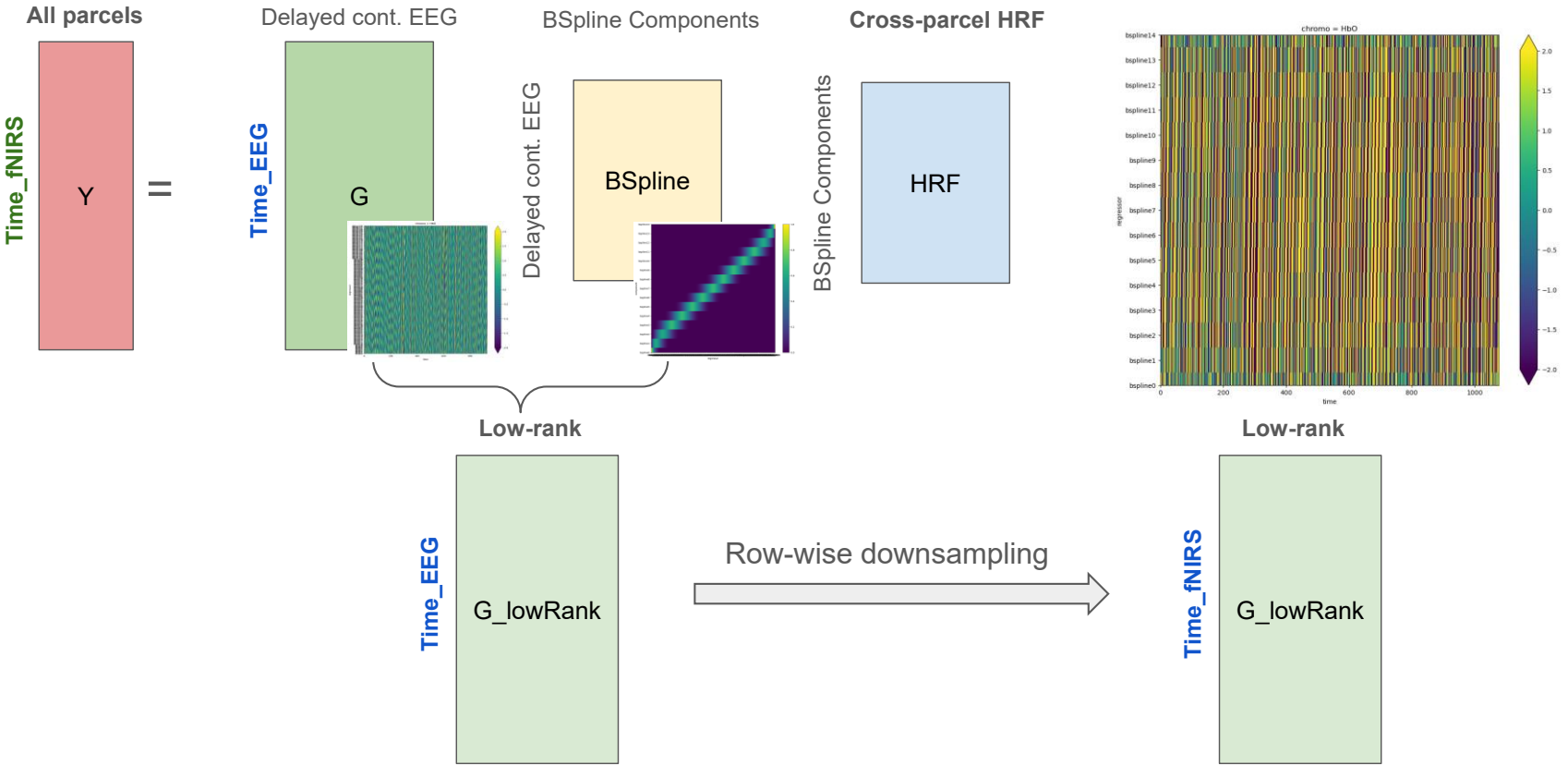


Example parcel-wise shared HRF

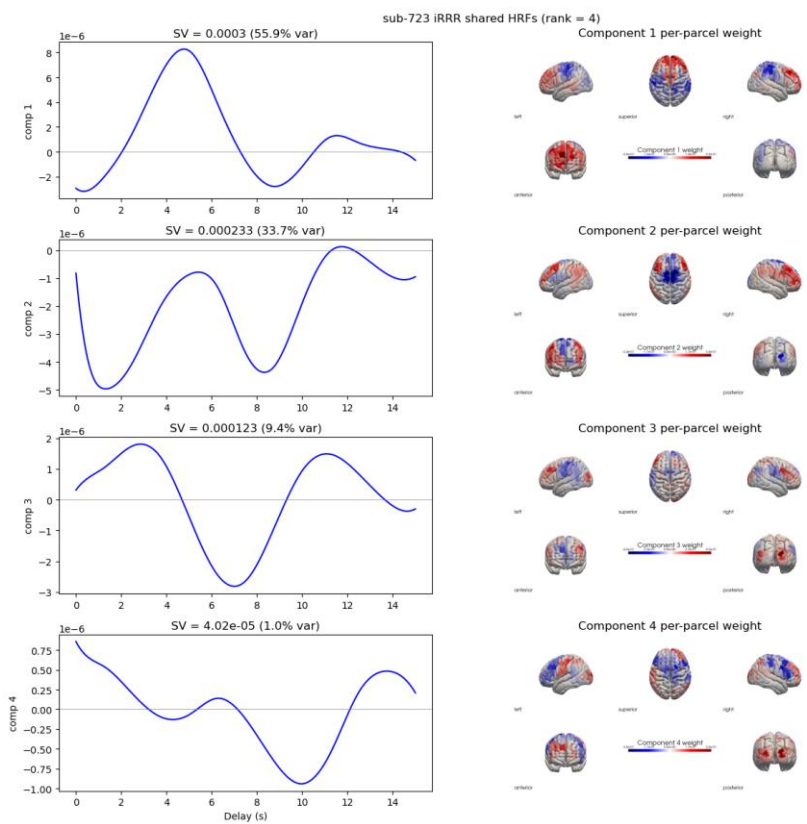


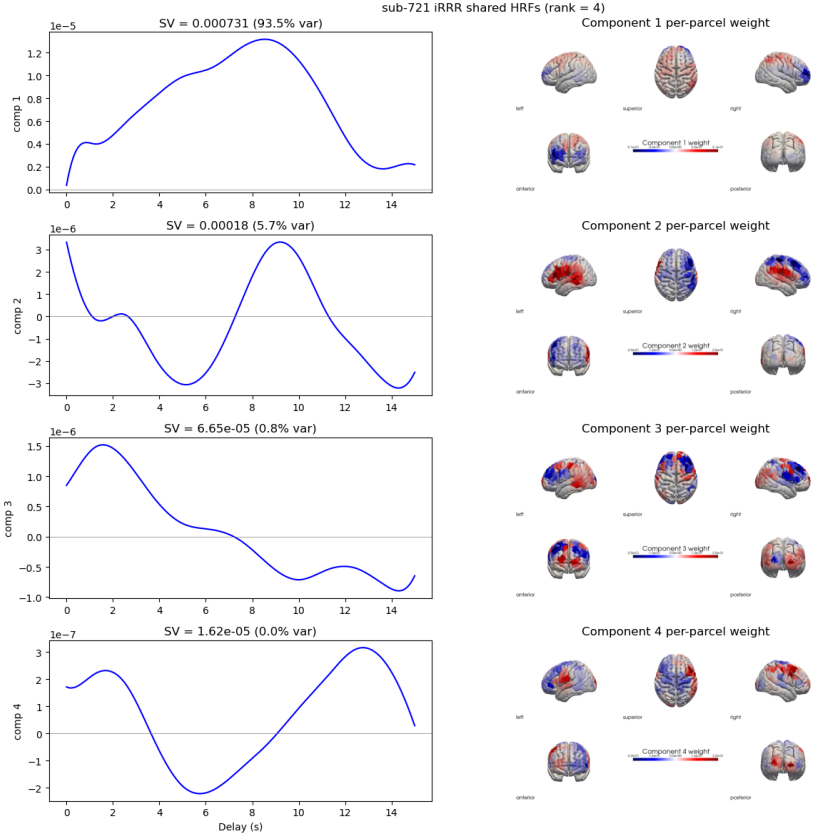


Cross-parcel HRF modeling using iRRR (actually RRR)



Example parcel-wise shared HRF found by iRRR



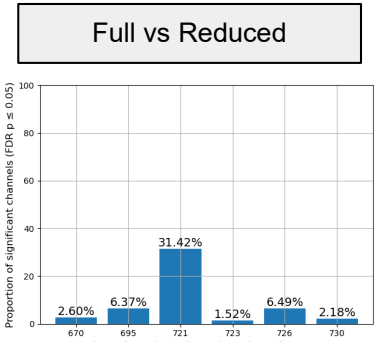


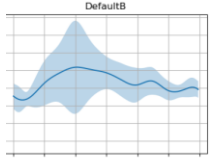
Take away

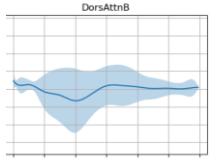
- Modulating event-related regressors with EEG ERP amplitude provide little additional information against directly using event-related regressors.

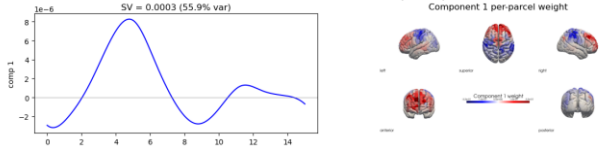
- The low-rank model shows that there are HRFs related to continuous EEG which is independent to events.

- The results of iRRR model indicate that there might be some HRFs which are shared across parcels.









Thank you